XAG329A

Agentic AI Course Learner Guide

Your 90 day journey at a glance, and how to move through all three parts.

90 DAYS · 3 PARTS · ABOUT 28.5 HOURS · SELF PACED

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What This Course Covers

By the end of the course, you will be able to do the following.



Design Agentic Systems

Handle complex, multi step, real world tasks with confidence.



Apply Test Time Scaling

Improve LLM performance by using inference time compute wisely.



Explain Self Improvement

Verifiers, feedback loops, reinforcement learning, and search.



Evaluate Tool Using Agents

Understand how agents take action and call external tools.



Extend Agents with Memory

Add retrieval and long term memory beyond the context window.



Build Robust Evaluations

Design frameworks that measure real agent performance.

How the Course Runs

Built for scale and autonomy, not around instructor availability.



One Place for Everything

Access Part 1, 2, and 3, the discussion forums, and live session schedules all from your main portal page.



Rolling Enrollment

Start any day. No cohort dependencies and no waitlists to join.



Fully Asynchronous

Move at your own pace. Knowledge checks and exercises inside the modules are practice and give you immediate feedback. They are not scored, but you do need to submit them, since course completion depends on it.



Two Live Sessions, Pacific Time

Sessions run on a rolling cadence, so whenever you enroll, two of them fall inside your 90 days. Session A covers what is new in agentic AI, including recent models, tools, and industry practice. Session B is applied, working through real examples and your questions about design and implementation.

Attendance is optional and recordings are posted on your portal.



Teaching Team Support

Team support is available via “Need Help” in Moodle.

The Full 90 Day Course Flow

Three self paced sprints. Each one builds on the part before it.

1

Part 1 · About 6 Hours · Final Exam

LLM and Agent Overview

Suggested: Days 1 to 15

2

Part 2 · About 10 Hours · Final Assignment

AI Agent Test-Time Scaling and Self-Improvement Techniques

Suggested: Days 16 to 45

3

Part 3 · About 12.5 Hours · Final Assignment

Agent Tool-Use, Memory, Planning/Reasoning, and Robust Evaluations

Suggested: Days 46 to 90

90 Days Is a Hard Cutoff

Once your 90 days end, you lose access to the platform, including any parts you have not finished. Submit your Part 3 assignment by Day 80 so there is time for grading before access ends.

Certification and Grading

Pass all three parts to earn your certificate.

85%+ PART 1 EXAM	**80%+** PART 2 CODING ASSIGNMENT
80%+ PART 3 CODING ASSIGNMENT	**2x/wk** GRADING CYCLE FOR ASSIGNMENT



Coding Assignments Are Graded Twice a Week

Part 2 and Part 3 assignments are graded on a twice weekly cycle, with results posted in Moodle.



Your Certificate

Pass Part 1, Part 2, and Part 3, and a Certificate of Completion is issued after your final result is posted.

What You Should Know

Four things to know up front. Nothing here needs doing on day 1.

What this course assumes you already know

Before you start, you should already have Python programming experience, some familiarity with LLMs including how to call an LLM API, and a basic understanding of machine learning concepts. Python matters most in Parts 2 and 3, because each one ends with a graded coding assignment in Python and Jupyter. Part 1 has no coding, but its modules on post training and reinforcement learning still assume that background.

Additional cost of $50 to $60, paid directly to OpenRouter

The Part 2 and Part 3 assignments call language models through OpenRouter, a paid API gateway. It is pay as you go, so there is no fixed fee and nothing to commit up front. Add a small amount of credit, around $10, and top up when it runs low. Across both assignments, expect $50 to $60 in total. You pay OpenRouter directly, it is separate from your course tuition, and it is not refundable through Stanford Online.

Accounts you will need, and when

Parts 2 and 3 require an OpenRouter account, a GitHub account, and a few free API accounts. You do not need any of them during Part 1, which needs nothing beyond your Moodle access. Each part guide lists exactly what that part needs, and repository access and setup steps arrive at the start of the part.

Third party tools, and where to get help with them

Parts 2 and 3 use services that Stanford Online does not own or operate. These include OpenRouter, Hugging Face, GitHub, and the free APIs used in Part 3. Stanford Online is not affiliated with them and does not endorse them, and your account, your billing, and your usage limits are between you and the provider. If a key will not generate, a payment fails, or a service is down, contact that provider's support. The teaching team can help with the course and the assignments, but not with problems inside another company's product.

Your Key Dates Worksheet

Fill this in once you enroll, then add each date to your personal calendar.

Enrollment (Day 1)

The day your 90 day clock starts.

Part 1 target (Day 15)

Enrollment date plus 15 days.

Part 2 target (Day 45)

Enrollment date plus 45 days.

Part 3 assignment submitted by (Day 80)

Early enough for grading, and for one resubmission if you need it.

Part 3, Day 90 Cutoff

Enrollment date plus 90 days. Access ends this day.

Live Session A (Pacific Time)

Announced on your portal once scheduled.

Live Session B (Pacific Time)

Announced on your portal once scheduled.

NOTES

Ready to Start?

Open Part 1 and start with Module 0, the Course Welcome. Then move into Module 1, meet your instructors, and let the 90 day journey begin.

Each part has its own guide covering the modules, the pacing, and what its final assessment involves. Read each one when that part opens.

Questions along the way? The teaching team is available through "Need Help" in Moodle.